PRACTICE SHEET

1. From past experience it is known that an investor will invest in security A with a probability of 0.6, will invest in security B with a probability 0.3 and will invest in both A and B with a probability of 0.2. What is the probability that an investor will invest neither in A nor in B?
(a) 0.7 (b) 0.28
(c) 0.3 (d) 0.4

2. Five coins whose faces are marked 2, 3 are thrown. What is the probability of obtaining a total of 12?
(a) 1/16 (b) 3/16
(c) 5/16 (d) 7/16

3. A coin is tossed three times. What is the probability of getting head and tail (HTH) or tail and head (THT) alternatively?
(a) 1/4 (b) 1/5
(c) 1/6 (d) 1/8

4. The probability that a student passes in mathematics is 4/9 and that he passes in physics is 2/5. Assuming that passing in mathematics and physics are independent of each other, what is the probability that he passes in mathematics but fails in physics?
(a) $\frac{4}{15}$ (b) $\frac{8}{45}$
(c) $\frac{26}{45}$ (d) $\frac{19}{45}$

5. From a pack of 52 cards, two cards are drawn, the first being

10. In a lottery, 16 tickets are sold and 4 prizes are awarded. If a person buys 4 tickets, what is the probability of his winning a prize?
(a) $\frac{4}{16^4}$ (b) $\frac{175}{256}$
(c) $\frac{1}{4}$ (d) $\frac{81}{256}$

11. If A and B are any two events such that $(A \cup B) = 3/4$, $P(A \cap B) = 1/4$ and $P(\bar{A}) = 2/3$, where $\bar{A}$ stands for the complementary event of A, then what is P(B)?
(a) 1/3 (b) 2/3
(c) 1/9 (d) 2/9

12. A card is drawn from a pack of 52 cards and a gambler bets that it is a spade or an ace. Which one of the following are the odds against his winning this bet?
(a) 13 to 4 (b) 4 to 13
(c) 9 to 4 (d) 4 to 9

13. A can hit a target 4 times in 5 shots:
B can hit a target 3 times in 4 shots:
C can hit a target 2 times in 3 shots:
All the three fire a shot each. What is the probability that two shots are at least hit?
(a) 1/6 (b) 3/5
(c) 5/6 (d) 1/3